

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

25th November 2020

Subject CS1A – Actuarial Statistics (Paper A)

Time allowed: 3 Hours 30 Minutes (9:30 – 13:00 Hours)

Total Marks: 100

- Q. 1)** The marks of a professional exam are normally distributed with unknown mean μ and standard deviation $\sigma = 20$.

Prior belief is that μ follows Normal Distribution with mean 65 and standard deviation 17.

- i) Calculate the prior probability that μ is greater than 60 (2)

Basis the sample of 150 scripts, mean marks are found to be 63.

- ii) Calculate the posterior probability that μ is greater than 60 (4)

- iii) Comment on your calculations from parts (i) and (ii). (2)

[8]

- Q. 2)** i) Given a Weibull Distribution with probability distribution function: $F(x) = 1 - e^{-(x)^2}$ determine the equation that needs to be solved to simulate a random variable using random number u from a $U(0,1)$ interval.

a) $x = 1 - e^{-(u)^2}$

b) $x = \sqrt{-\ln(1 - u)}$

c) $x = -\ln(\sqrt{1 - u})$

d) $x = \sqrt{-\ln(1 - u^2)}$ (2)

- ii) Simulate a value at $u=0.75$ based on your answer in part (i) (1)

- iii) Given a probability density function $f(x) = (1 + x)^{-2}$ for $x > 0$ determine the corresponding probability distribution function

a) $F(x) = 1 - (1 + x)^{-1}$

b) $F(x) = (1 + x)^{-1}$

c) $F(x) = 1 - (1 + x + x^2)^{-1}$

d) $F(x) = 1 - (1 + x)^{-3}$ (2)

- iv) Hence simulate a random variable from the distribution function as per your answer in part (iii) using random variable $u=0.5$ from $U(0,1)$ (2)

[7]

- Q. 3)** Answer the following in case of a bivariate dataset with data points (X_i) and (Y_i) :

- i) Identify the correct definition of the Spearman rank correlation coefficient

$$\begin{aligned}
 \text{a)} \quad r_s &= 1 - \frac{6 \sum_i d_i^2}{n(n^2+1)} \\
 \text{b)} \quad r_s &= 1 - \frac{6 \sum_i d_i^2}{n^2(n-1)} \\
 \text{c)} \quad r_s &= 1 - \frac{6 \sum_i d_i^2}{n(n-1)} \\
 \text{d)} \quad r_s &= 1 - \frac{6 \sum_i d_i^2}{n(n^2-1)} \tag{1}
 \end{aligned}$$

ii) Define the symbols in your answer for part (i) and the condition that is necessary so that the formula can be used to calculate the respective coefficients. (1)

iii) Define concordant pair of observation (1)

iv) Identify the correct definition of the Kendall rank correlation coefficient

$$\begin{aligned}
 \text{a)} \quad \tau &= \frac{n_c - n_d}{n(n-1)} \\
 \text{b)} \quad \tau &= \frac{n_c - n_d}{n(n-1)/2} \\
 \text{c)} \quad \tau &= \frac{n_c - n_d}{n^2} \\
 \text{d)} \quad \tau &= \frac{(n_c - n_d)/2}{n(n-1)} \tag{1}
 \end{aligned}$$

v) Define the symbols in your answer for part (iv) and the condition that is necessary so that the formula can be used to calculate the respective coefficients. (2)
[6]

Q. 4) Let p denote the proportion of policies where claim is registered. Prior belief about p is that it follows Beta distribution with parameters α and β . Underwriters estimate mean μ and standard deviation as σ of p .

i) Select how α and β would be represented in terms of μ and σ (3)

$$\begin{aligned}
 \text{a)} \quad \alpha &= \frac{\mu^2(1-\mu) - \mu \sigma^2}{\sigma^2} \text{ and } \beta = \frac{(\mu(1-\mu) - \sigma^2)(1-\mu)}{\sigma^2} \\
 \text{b)} \quad \alpha &= \frac{\mu(1-\mu) - \sigma^2}{\sigma^2} \text{ and } \beta = \frac{(\mu(1-\mu) - \sigma^2)(1-\mu)}{\mu \sigma^2} \\
 \text{c)} \quad \alpha &= \frac{\mu(1-\mu) + \sigma^2}{\sigma^2} \text{ and } \beta = \frac{(\mu(1-\mu) + \sigma^2)(1-\mu)}{\mu \sigma^2} \\
 \text{d)} \quad \alpha &= \frac{\mu^2(1-\mu) + \mu \sigma^2}{\sigma^2} \text{ and } \beta = \frac{(\mu(1-\mu) + \sigma^2)(1-\mu)}{\sigma^2}
 \end{aligned}$$

A random sample of n policies is considered, and it is observed that d claims arise of them.

ii) Derive the posterior distribution of p (3)

- iii) If the mean of posterior distribution is written in the form of a credibility estimate, select the credibility factor, Z :

a) $Z = \frac{\alpha}{(\alpha + \beta + n)}$

b) $Z = \frac{n\alpha}{(\alpha + \beta + n)}$

c) $Z = \frac{(\alpha + \beta)}{(\alpha + \beta + n)}$

d) $Z = \frac{n}{(\alpha + \beta + n)}$

(3)

- iv) Select which of the below statement is true:

a) MLE of p is d/n ; Z increases with increasing σ^2

b) MLE of p is n/d ; Z increases with increasing σ^2

c) MLE of p is d/n ; Z decreases with increasing σ^2

d) MLE of p is n/d ; Z decreases with increasing σ^2

(3)

- v) Comment on the chosen option regarding the dependency of credibility factor on σ^2 from part (iv)

(1)

[13]

- Q. 5)** A general insurance company is studying a portfolio of policies on which the company has received more than 1 claims. The analyst has generated the following frequency table:

Number of claims	Frequency
2	230
3	54
≥ 4	6

It is believed that the claims follow a Poisson distribution with parameter λ (for ease of typing, you may write the parameter as 'L' in your solution)

- i) Show that the truncated probability function is given by:

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x! (1 - e^{-\lambda} - \lambda e^{-\lambda})} \quad (4)$$

- ii) Prove that the relation between sample mean $\bar{x}$ and λ is given by

$$\bar{x} = \frac{\lambda(n-1)(1 - e^{-\lambda})}{n(1 - e^{-\lambda} - \lambda e^{-\lambda})}$$

(Hint: You can use method of maximum likelihood to find the answer)

(5)

- iii) Using interpolation and guess value of λ as 0.6 and 0.7, calculate the estimate of λ upto 3 decimal places.

(4)

[13]

- Q. 6)** The country of Actuaria has been affected by some novel virus species. The health department has compiled the following data as regards the travel history and symptoms seen in some of the people of Actuaria that were tested for the virus:

	No Symptoms (NS)	Some Symptoms (SS)	Confirmed case (CC)
Travelled to place outside Actuaria (T)	86	25	80
No travel in recent past (NT)	130	148	31

Test at 1% level if there is any association between travel history and the symptoms indicating presence of virus in the body.

[8]

- Q. 7)** i) Variance of random variable X in terms of probability generating function is given as:

a) $var(X) = G_X''(0) + G_X'(0) - (G_X'(0))^2$

b) $var(X) = G_X''(1) - G_X'(1) + (G_X'(1))^2$

c) $var(X) = G_X''(1) + G_X'(1) - (G_X'(1))^2$

d) $var(X) = G_X''(0) - G_X'(0) + (G_X'(0))^2$

(1)

- ii) The infinite series : $1 + tE(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \dots$ is expanded form of...

a) Probability Generating Function

b) Moment Generating Function

c) Cumulant Generating Function

d) None of the above

(1)

A person is observing time taken for a bus to appear at the bus stop. It is believed that the time period (denoted by random variable X) between the time at which two consecutive buses arrive at the stop follows an exponential distribution with mean θ .

- iii) The following equation on further evaluation yields the moment generation function of X.

$$\begin{aligned}
 \text{a) } E(e^{tX}) &= \frac{1}{\theta} \left(\frac{1}{\theta} - t \right) \int_0^\infty \exp(-z) \cdot dz \\
 \text{b) } E(e^{tX}) &= \frac{1}{\theta} \int_0^\infty \exp(-z) \cdot dz \\
 \text{c) } E(e^{tX}) &= \left(\frac{1}{\theta} - t \right)^{-1} \int_0^\infty \exp(-z) \cdot dz \\
 \text{d) } E(e^{tX}) &= \frac{1}{\theta} \left(\frac{1}{\theta} - t \right)^{-1} \int_0^\infty \exp(-z) \cdot dz
 \end{aligned} \tag{2}$$

iv) If random variable Y denotes the total observation time to count N buses, determine the moment generating function of Y. (3)

v) Hence identify the distribution followed by Y. (2)

[9]

Q. 8) The theme park management wants to study the relationship between the number of visitors and temperature of the day. In the sample collated for 25 low temperature weeks and 35 high temperature weeks over the span of 15 months, x_i denotes the number of visitors in week i . Given below is the summary of number of visitors:

Type of week	$\sum X_i$	$\sum X_i^2$
Cold week	10,250	4,718,190
Warm week	12,001	4,740,201

i) Perform a test with the null hypothesis that the variance of number of visitors during cold weeks is equal to the variance of number of visitors in warm weeks against the alternative that the variance is higher in cold week when compared to warm week.

Significance level of 5% to be used. (5)

ii) Perform a test to check if there is any significant difference in the mean of the number of visitors to theme park during cold weeks and warm weeks.

Significance level of 5% to be used. (4)

To obtain further insight into the relationship between temperature and number of visitors, the management decides to carry out a linear regression analysis.

Let T denote the average temperature of the week and X denote the total number of weekly visitors. The model suggested is $E(X) = AT + B$. Given below is the summary for temperature t_i and visitors x_i for 78 weeks:

$$\sum t_i: 1,603$$

$$\sum t_i * x_i: 582,205$$

$$\sum t_i^2: 51,823$$

- iii) Estimate the correlation co-efficient $\rho(T,X)$ between the temperature and visitors. Comment on the same. (5)
- iv) Estimate the parameters A and B (2)
- [16]

Q. 9) It is known that for a shop of a particular category, the mean invoice amount towards sales is INR 2000 and standard deviation is INR 500. Stating clearly any assumptions that you make, calculate the following:

- i) Probability that mean amount of next 10 invoices is less than INR 1700. (3)
- ii) Probability that standard deviation of amount of next 10 invoices is less than INR 250. (3)
- iii) Probability that both (i) and (ii) occur together. (i.e. mean amount of next 10 invoices is less than INR 1700 and standard deviation of amount of these 10 invoices is less than INR 250) (1)
- [7]

- Q. 10) i)** Explain what is meant by a saturated model and discuss if such a model is useful in practice. (2)
- ii) a) Define both Pearson and deviance residuals (2)
- b) Explain how these two types of residuals are different (2)
- c) State in which case they are the same (1)

It is believed that the claims arising out of Health Insurance primarily depends on two variables – Age and BMI:

Age x_i is the category 18 to 35, 36 to 45, 46 to 60 and above 60

BMI y_i can be categorised as Under-weight, Normal, Over-weight and Obese depending on the BMI values

The company wishes to model claim amounts using this data and is investigating models which take into account the below variables:

Model	Choice of predictor	Scaled Deviance
A	1	900
B	Age	800
C	Age + BMI	770
D	Age * BMI	760

- iii) Carry out the analysis of scaled deviance and recommend which model should be used by the Insurance Company? (6)
- [13]
